自觉遵守考场纪律 如考试作弊 此答卷无效

姓名_____ 学号_____

密 封 线

东 南 大 学 考 试 试 卷（A 卷）

课程代码	BJSL0041	课程名称	离散数学	考试学期	2022-2023-3
适用专业	计算机类	考试形式	闭 卷	考试时长	120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

题目	1	2	3	4	5	6	7	8	9	10	总分
得分											
批阅人											

1. Choice questions（2 points for each question）

(1). Which of the following statement is NOT true? ( )

A. If 1=1, then 1=1;

B. π is a rational number, if $\pi > 3$;

C. 0=1, only if 1+1=2;

D. If 0=1, then x^2=-1 has real root.

(2). Which of the following formular pairs is NOT equivalence? ( )

A. $\neg(P \to Q)$ and $P \wedge \neg Q$;

B. $P \wedge Q$ and $\neg(\neg P \vee \neg Q)$;

C. $\neg(P \to \neg Q)$ and $P \wedge Q$;

D. $P \wedge Q$ and $\neg P \to Q$.

(3). Which of the following set statements is true? ( )

A. $|P(\varnothing)| = 2$.

B. $A - (B \times C) = (A - B) \times (A - C)$.

C. $A = A \cap (A \cup B)$

D. $|A| - |B| = |A - B|$.

(4). Determine which of these functions from R to R is surjective. ( )

A. f(x) = -3x+4　　B. f(x) = $-2x^2+4$

C. f(x) = e^x　　D. f(x) = $\lceil x/2 \rceil$

(5). A finite group G has a subgroup S with order |S|=3, the order of group G cannot be ( ).

A. 3 B. 5 C. 6 D. 9

(6). Let $< L, \wedge, \vee, ', 0, 1 >$ be a Boolean algebra, which of the following is not equivalent to the others? ( )

A. $(a' \vee b)' = 0$ B. $a \wedge b' = 0$

C. $(a \wedge b)' = b$ D. $a \vee b = b$

(7). Which graph that shows below is a bipartite? ( )

A. K_3.

B. C_5.

C. C_6.

D. W_6.

(8). If an simple undirected graph G=(V, E) contains 10 vertices and to ensure that the graph G is connected in any case, the minimum number of edges required is ( ).

A. 45 B. 36 C. 37 D. 39

2. Completion（2 points for each question）

(1). Let p: "There is rain" and q: "I buy an umbrella". For the statement of the statement "If there is rain, then I buy an umbrella", its converse statement is_____ _____ and its inverse statement is _____.

(1 point for each blank)

(2). Given P = {3, 4, 5, 6, 7}, the truth value of statement $(\exists x \in P)\ (x\ mod\ 3 = 1)$ is (true or false) and the value of x that make the proposition satisfiable is _____. (1 point for each blank)

(3). Let A = {a, b}, how many distinct equivalence relations are there on set A _____ , how many distinct partial orderings are there on set A _____.

(1 point for each blank)

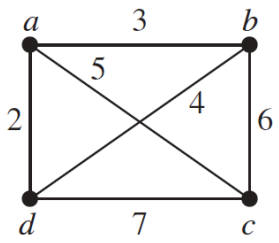
(4). Let G be a group and H is a subgroup of G, then for any $a, b \in G$, $Ha \cap Hb =$

_____.

(5). Let $(N_8, +_8)$ be the group under the operation of module 8 addition. If H is a subgroup containing 5, then $H =$_____.

(6). In an undirected graph G(V, E), |V| = n, |E| = m. The sum of the degrees of all vertices is equal to_____.

(7). Solve the traveling salesperson problem for the below graph by finding the total weight of all Hamilton circuits, and then determining a circuit with minimum total weight. That is_____.

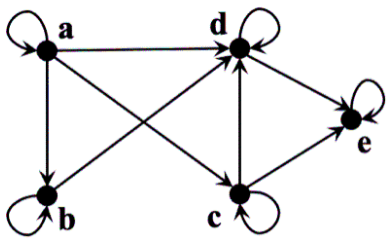


3. Compute the CNF(Conjunctive normal form) of $(\neg P \to R) \wedge (Q \leftrightarrow P)$ (7 points)

4. Build valid arguments for: (9 points)
Hypothesis: P ∧ S, P → ¬R, ¬R → ¬Q and conclusion: S ∧ ¬Q

5. Let R_1 and R_2 be relations on A, relation R={(a, b) | (a, b)∈ $R_1 \wedge (a, b) \in R_2$}, prove that R is an equivalence relation if R_1 and R_2 are equivalence relations. (9 points)

6. Let A={a, b, c, d, e}, R is a relation represented by the following directed graph.(9 points)



(1) Find the transitive closure t(R) of R.
(2) Determine whether the transitive closure t(R) is a partial order. If it is a partial order, draw the Hass diagram, and find all the upper bounds and lower bounds of {b, c, d}.

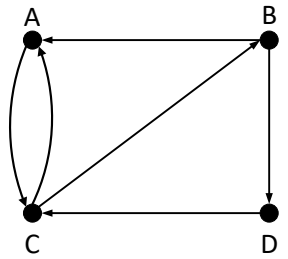
7. Let $(Z, +, \times)$ and $(N_4, +_4, \times_4)$ be two algebraic systems, where Z is the set of integers and $N_4 = \{0, 1, 2, 3\}$, $+, \times$ are normal integer addition and multiplication and $+_4$, $\times_4$ are the module 4 addition and multiplication. Let $f: Z \to N_4, f(a) = a \ (mod\ 4)$. (9 points)
(1) Is f a homomorphism? Justify your answer.
(2) Is f injective? Justify your answer.
(3) Is f surjective? Justify your answer.

8. Let $S = \{1, 2, 3, 6, 7, 14, 21, 42\}$. (9 points)

(1) Construct a Boolean algebra by defining three operations on S and find the greatest and least element.

(2) Show the above Boolean algebra is isomorphic to $(\mathcal{P}(S), \cap, \cup, \sim, \emptyset, S)$ with $S = \{a, b, c\}$.

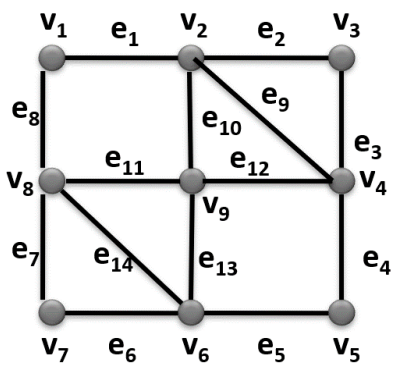
9. A directed graph G = (V, E) is given below: (9 points)



(1) Please give out the adjacency matrix of G.

(2) How many paths whose length is equal to 3?

(3) How many paths from *A* to *D* whose length is less than or equal to 4?

(4) How many circuits from A to A whose length is less than or equal to 4?

10. Please use *Fleury* algorithm to solve the Eulerian Circuits shown below. (9 points)